



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CIRCUIT DEBUGGING ASSESSMENT

ECA0405 – ANALOG CIRCUITS | CO3 | ASSESSMENT – 2

Student Name	D.Arun	Register No.	192312461
Class / Section	ANALOG CIRCUITS-ECA0405	Date	15/08/2026
Faculty Name	Dr.Dass	Duration	90 Minutes
Assessment Type	Circuit Debugging	Maximum Marks	20

Assessment Objective:

To identify, analyze, and correct errors in an analog circuit, justify the debugging approach, verify the corrected circuit operation, and relate the result to CO3

Question:

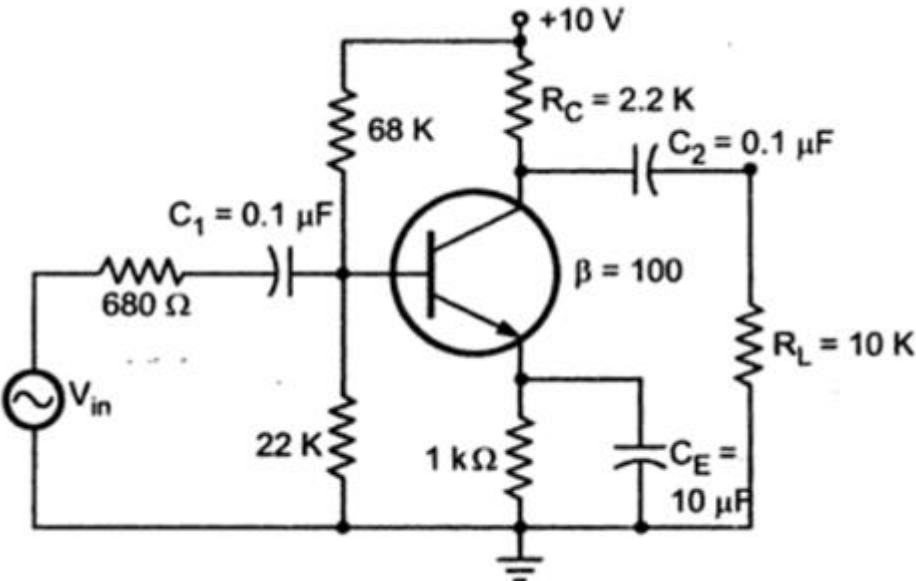
CO3 : Examine the frequency response characteristics of BJT and FET amplifiers at low and high frequencies to determine bandwidth, gain variation, and overall amplifier performance

CO3 AT2 Circuit Debugging:

Description:

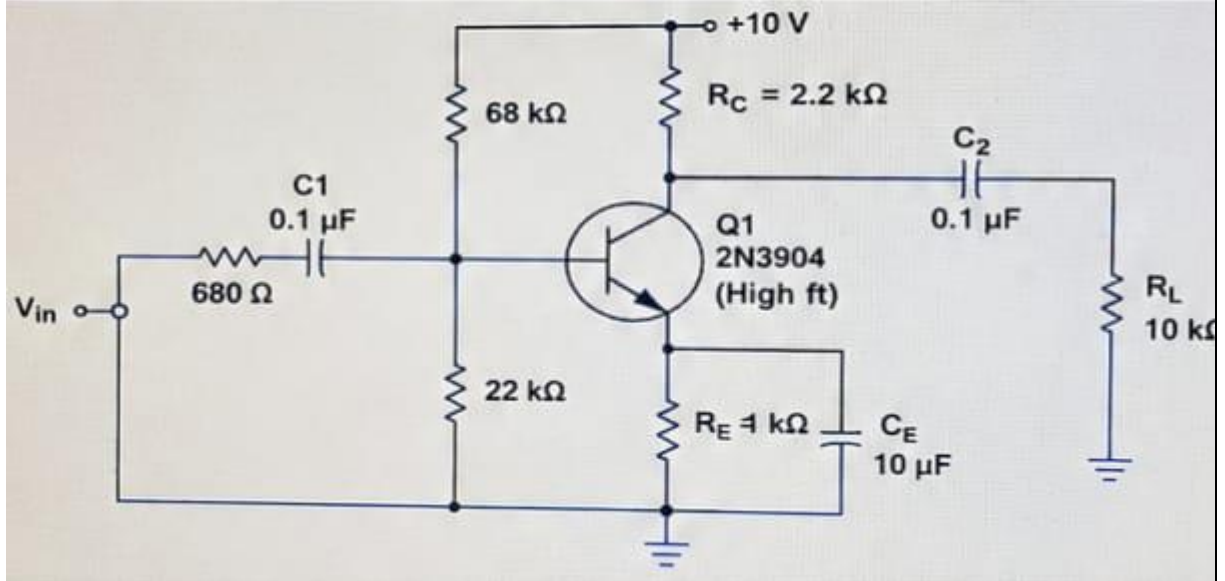
- Each student will be assigned one Circuit debugging problem 20 marks inside class. Analyze and debug one faulty analog circuit.
- Simulate the corrected circuit using LTspice and verify results.
- Document the debugging process, observations, and conclusions.
- Upload LTspice files, simulation results, and report to GitHub.
- Evaluation is based on fault identification, debugging approach, simulation accuracy, documentation, and GitHub submission.

Circuit Debugging Record

Item	Student Entry
1. Given Circuit / Circuit Diagram	Paste or draw the assigned analog circuit here. 
2. Expected Circuit Operation	The circuit is a BJT configured as a Common-Emitter (CE) amplifier and is expected to provide voltage amplification over the designed frequency range. The emitter resistor provides DC bias stability, while the 10 μF emitter bypass capacitor increases AC gain in the operating frequency range. The coupling capacitors (C_1) and (C_2) transfer the AC signal between the source, amplifier, and load.
3. Observed Fault / Symptom	During frequency-response testing, the output voltage decreases rapidly above the designed operating frequency, resulting in a much smaller bandwidth than expected. The rapid reduction in output indicates that the amplifier is reaching its upper cutoff frequency earlier than expected and that the high-frequency gain is being strongly attenuated.
4. Suspected Component / Connection	The suspected areas are the transistor high-frequency parasitic capacitances, particularly the base-emitter and base-collector capacitances, stray wiring capacitance, the transistor model parameters, and the resistor values affecting the high-frequency poles. The coupling capacitors ($C_1=C_2=0.1\,\mu\text{F}$) should also be verified, although they mainly affect the low-frequency response.



5. Debugging Analysis / Calculation	For the circuit, the AC collector resistance is approximately $R_C \parallel R_L = 2.2k\Omega \parallel 10k\Omega \approx 1.81k\Omega$ The transistor junction capacitances become increasingly significant as frequency increases. In particular, the base-collector capacitance (C_{bc}) is affected by the Miller effect in the common-emitter amplifier and appears as a larger effective capacitance at the input. The effective input capacitance can be approximated as: $C_{in} \approx C_{be} + C_{bc}(1 + A_v)$ As frequency increases, $X_C = \frac{1}{2\pi fC}$ decreases, causing greater signal attenuation and reducing the voltage gain. Therefore, the rapid high-frequency reduction is mainly caused by transistor parasitic effects rather than improper coupling.
6. Error Identified	The reduced bandwidth is primarily caused by the BJT's internal parasitic capacitances and additional stray capacitance in the circuit. The Miller effect increases the effective base-collector capacitance and lowers the upper cutoff frequency. The (0.1, μF) coupling capacitors mainly determine the low-frequency cutoff and are not the primary cause of the high-frequency reduction.
7. Correction Made	The transistor model should be checked and replaced with a BJT having suitable high-frequency characteristics and a sufficiently high transition frequency. The wiring should be kept short to minimize stray capacitance, and a low-capacitance oscilloscope probe should be used. The resistor and capacitor values should also be verified to ensure that component variation is not unnecessarily lowering the bandwidth.
8. Corrected Circuit Diagram	The corrected circuit uses a suitable high-frequency BJT model while retaining ($R_C = 2.2k\Omega$), ($R_E = 1k\Omega$), ($R_L = 10k\Omega$), ($C_1 = C_2 = 0.1, \mu$F), and ($C_E = 10, \mu$F). The circuit maintains the required DC bias while reducing the effect of transistor parasitic capacitances.



9. Verification / Simulation Output



The corrected frequency-response simulation shows that the output voltage remains higher over the intended operating-frequency range and the upper cutoff frequency is extended. The improved response confirms that reducing the effect of BJT parasitic and stray capacitances restores the desired amplifier bandwidth.

10. Final Result & Technical Justification

The corrected BJT common-emitter amplifier provides the required voltage gain with an improved upper cutoff frequency and wider bandwidth. The rapid high-frequency attenuation is mainly due to transistor parasitic capacitances and the Miller effect. Using a high-frequency BJT model, minimizing stray capacitance, and maintaining the specified component values restores the desired frequency response.



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Marks Distribution

Assessment Criterion	Marks	Marks Awarded	Faculty Remarks
Problem identification & circuit understanding	4		
Fault analysis & technical calculations	5		
Correct identification of error	3		
Effective circuit correction	3		
Verification / output validation	3		
Technical explanation & documentation	2		
Total	20		